What a causal mask is worth: a rank–logit law for prefix averaging*

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Abstract

Transformer models use causal masks to force each position to assign zero attention weight to future tokens. We ask what it costs for an unmasked dense attention mechanism to approximate this behavior by giving sufficiently little weight to those future tokens. We study this question in prefix averaging with a single, position-dependent unmasked head. To achieve polynomial accuracy, the logits required become extremely large. The smallest centered logit radius B is governed by the law below, where $p > 1$ and positive integer r are fixed, score matrices have centered score rank at most r and worst-row output error is at most n^{-p}:

$$\log B = \frac{n-1}{r}\left((p-1)\log n + \log\log n + O_{p,r}(1)\right).$$

Matching lower and upper bounds show that this formula gives the optimal asymptotic cost at every fixed polynomial error target n^{-p}, including both diverging terms. The lower bound uses a determinant argument, while the upper bound constructs a balanced rank-r matrix. We also compare the construction with these bounds numerically at finite sequence lengths. A separate estimate shows that when $p > 3/2$, restricting the centered logit radius to polynomial size forces the rank to grow proportionally with sequence length.

These large logits force the product of the maximum query norm and maximum centered-key norm to be large. For example, there are no query and key coordinates inside float32's finite range that would achieve error n^{-2} with a position-indexed pure dot-product head with dimension 128 once $n \geq 4166$. The same fixed-rank law extends to a content-adaptive model in which the Boolean values being averaged also affect the queries and keys.

1 Introduction

Attention is often described as a learned mechanism, where queries and keys allow tokens to attend and influence one another. However, architectural mechanisms also determine which key positions each query is allowed to attend to before these scores are passed through softmax. In autoregressive transformers, a causal mask supplies one such rule by preventing every position from attending to future tokens. Causal masking is standard in autoregressive transformers (Vaswani et al., 2017). To our knowledge, the literature has not yet quantified the cost of reproducing the same operator without the mask.

We want to understand how much representational work this causal mask performs. We do this by removing the causal mask and asking a dense attention head to reproduce the same behavior using only its score matrix. Measuring the additional score rank and logit magnitude required by an unmasked head allows us to quantify the contribution of the causal mask, and also separates structure supplied directly by architecture from structure that is represented through learned parameters.

We study this question in the setting of prefix averaging, a standard primitive in theoretical transformer constructions (Yang et al., 2026; Strobl et al., 2025). In prefix averaging, a causal mask creates a moving

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support boundary where all tokens after the current position are masked and receive zero weight, while all visible tokens receive equal weight. Let

$$T(i,j) = \frac{1}{i}\mathbf{1}[j \le i]$$

be the prefix-average operator. With causal masking, we can achieve T exactly. With the mask removed, softmax forces us to assign positive mass to every key. To approximate T with unmasked softmax, we'll require nearly equal scores on each prefix along with strong suppression of every future key.

For an unmasked score matrix S, define $P_0 = I_n - \mathbf{1}\mathbf{1}^\top/n$, so that SP_0 subtracts the average from each row of S. The centered score rank is $\rho(S) = \mathrm{rank}(SP_0)$, and the centered logit radius is $B(S) = \max_{i,j} |(SP_0)(i,j)|$. For a pure dot-product head of dimension d_h, $\rho(S) \le d_h$. Because softmax is unchanged by row shifts, $\mathrm{softmax}(S) = \mathrm{softmax}(SP_0)$. If $\rho(S) \le r$, then SP_0 admits a factorization $QK^\top$ using r-dimensional query and key vectors. If two finite logit rows produce the same softmax probabilities, they can only differ up to rowwise constants. Therefore, $\rho(S)$ is the smallest dimension needed to factorize any score matrix that produces the same attention pattern. For every fixed $p > 1$ and fixed positive integer r, if B denotes the minimum centered logit radius required to achieve worst-row output error n^{-p}, our main theorem gives

$$r \log B = (n-1)\left((p-1)\log n + \log\log n + O_{p,r}(1)\right)$$

as the optimum for error n^{-p}.

This result says that a fixed-rank unmasked head must use centered logits whose magnitude increases extremely quickly with sequence length. In this problem, rank and logit magnitude can substitute for one another: extreme scores allow a fixed-rank softmax to approximate the boundary, while higher rank supplies more independent score patterns with which to represent the changing boundary. A second result states that if the centered logit radius is restricted to polynomial size and $p > 3/2$, then the rank must grow proportionally with sequence length.

To achieve our main result, we use a determinant argument to show that no rank-r score matrix can achieve the target with substantially smaller logits. For the matching upper bound, we construct a balanced rank-r matrix that achieves the same asymptotic scale.

We checked our result computationally. At selected values of n and r, we compare the realized centered logit radius of our construction with the determinant lower bound and the construction's closed-form upper budget. The construction lies between these bounds in every tested case. A factor-norm inequality then converts the logit requirement into a lower bound on the product of the maximum query norm and maximum centered-key norm. For example, no position-indexed pure dot-product head of dimension 128 whose query and key coordinates lie within float32's finite range can achieve error n^{-2} once $n \ge 4166$. Under the same assumptions, head dimensions through 252 are also insufficient once $n \ge 8192$.

The results above hold one score matrix fixed while the value stream varies. We next consider a restricted adaptive model in which the Boolean values being averaged also affect the queries and keys. Uniform accuracy over all Boolean inputs forces the attention matrix on the all-zero input to approximate causal prefix attention, so the same fixed-rank law holds in this model.

To our knowledge, this is the first quantitative law for the rank–logit cost of replacing a causal mask with dense unmasked attention in prefix averaging. Our results suggest that structural encoding is a powerful representational resource. When we know in advance that certain token pairs shouldn't interact, explicitly masking those pairs may reduce the score rank or logit magnitude that dense attention would otherwise require to approximate zero attention. Our result also shows that rank alone can give an incomplete picture of representational cost, as a low-rank head may be expressive enough while requiring massive score magnitudes that are numerically unrepresentable as sequence length grows.

The remainder of the paper is organized as follows. Section 2 discusses related work. Section 3 sets up our problem with formal definitions for our key mathematical objects. Section 4 presents our main rank–logit law and the resulting rank–magnitude tradeoffs. Section 5 explains the determinant mechanism behind the lower bound and the matching rank-r construction. Section 6 presents a finite-size numerical audit that brackets the optimum and derives bounds on query and key magnitudes. Section 7 extends the law to heads whose queries and keys depend on the Boolean payload being averaged. Section 8 states three open questions. The appendices contain the complete proofs and experimental details.

2 Related work

Transformer models combine learned mechanisms with architectural decisions. In the original Transformer decoder, future positions are masked to preserve the autoregressive property (Vaswani et al., 2017). This represents a different source of structure from learned dot-product scores.

Construction-oriented work uses causal masking to obtain prefix means or prefix-sum primitives (Yang et al., 2026; Strobl et al., 2025); our result quantifies the cost of replacing that masked primitive with one fixed-rank dense softmax head.

Prior literature focuses on the behavior of the learned score mechanism. Bhojanapalli et al. (2020) show that head dimension can create a low-rank bottleneck that prevents a head from representing some attention matrices. Amsel et al. (2025) show, for a nearest-neighbor target, that adding many low-rank heads need not compensate for insufficient rank. Likhosherstov et al. (2023) show that sparse attention matrices can be approximated with query/key dimension logarithmic in sequence length when the error and sparsity parameters are fixed. These papers ask how constraints on rank or dimension affect which attention patterns are representable. We instead look at one attention pattern and study how large the logits need to become to approximate it at fixed centered rank.

Veličković et al. (2025) show that when logit spread stays bounded, each global softmax coefficient disperses on the order of $1/n$ as the number of items grows. Our result quantifies the logit magnitude required to avoid such dispersion and relates it to score rank for causal prefix averaging.

Lee (2026) studies row-centered logits as well under the name “energy field,” and shows that tested trained models concentrate variance in relatively few components. This suggests that a version of our result might eventually be relevant to trained heads that behave approximately like low-rank matrices. This remains a hypothesis.

Alman & Song (2023) show that the time required to approximate dense attention changes with bounds on the input entries. They ask how quickly an attention output can be computed from given query and key factors; we ask how large those factors must become to represent one target attention pattern at fixed rank.

3 Problem setup

Let S denote the complete pre-softmax score matrix, including any additive positional bias, and let $A = \mathrm{softmax}_{\mathrm{row}}(S)$. For a pure dot-product head, we write

$$S(i,j) = \langle q_i, k_j \rangle,$$

with the conventional $1/\sqrt{d_h}$ scale absorbed into the factors. All rank and radius bounds below apply to the complete matrix S. The relation $\rho(S) \le d_h$ holds for pure dot-product scores; a separately parameterized positional bias can add rank independently of d_h. Softmax is invariant under row shifts, so SP_0 is the identifiable score matrix. Its radius controls every within-row score difference:

$$|S(i,j) - S(i,t)| \le 2B(S).$$

For a bounded value stream v, the output is Av. The operational error is

$$\sup_{\|v\|_\infty \le 1} \|(A-T)v\|_\infty = \max_i \|A(i,:) - T(i,:)\|_1. \tag{1}$$

Hölder’s inequality gives the upper bound, and the sign vector of the largest-error row gives equality. The metric asks one realized attention matrix to average every bounded value stream.

Uniformity ranges over separate value streams while A stays fixed. A single fixed value matrix tests the column span that it presents to the head.

The radius $B(S)$ also admits an architectural bound.

Proposition 1 (Factor-norm bound). *Let $S = QK^\top$ with rows $q_i, k_j \in \mathbb{R}^{d_h}$, and let $\bar{k} = \frac{1}{n}\sum_j k_j$. Then*

$$B(S) \le \max_i \|q_i\|_2 \cdot \max_j \|k_j - \bar{k}\|_2. \tag{2}$$

If in addition every entry of Q and K has magnitude at most F, then $B(S) \le 2d_h F^2$. Consequently, if $B(S) > 0$ and $F > 0$, then $\log B(S) \le \log 2 + \log d_h + 2\log F$.

Proof. Centering a score row is equivalent to centering the keys: row i of SP_0 has entries $\langle q_i, k_j\rangle - \frac{1}{n}\sum_t \langle q_i, k_t\rangle = \langle q_i, k_j - \bar{k}\rangle$, and Cauchy–Schwarz gives (2). Under the entrywise bound, $\|q_i\|_2 \le \sqrt{d_h}F$ and $\|k_j - \bar{k}\|_2 \le 2\sqrt{d_h}F$, hence $B(S) \le 2d_h F^2$. Taking logarithms gives the final claim when the quantities are positive. □

Thus every lower bound on $B(S)$ is also a lower bound on $\max_i \|q_i\|_2 \max_j \|k_j - \bar{k}\|_2$. If the conventional $1/\sqrt{d_h}$ scale is written explicitly rather than absorbed into the factors, the right-hand side of (2) is divided by $\sqrt{d_h}$.

Write $[m] = \{1, \dots, m\}$. For a finite set, osc denotes the maximum minus the minimum, and $\left\{ {m \atop k} \right\}$ denotes a Stirling number of the second kind. Let $I \subseteq [n-1]$ and put

$$I^+ = I \cup \{j + 1 : j \in I\}.$$

Row $i \in I$ has a causal (δ_i, γ_i) profile *on* I when

$$\operatorname{osc}_{j \in I^+,\ j \le i} S(i, j) \le \delta_i, \qquad S(i, i+1) \le S(i, i) - \gamma_i. \tag{3}$$

Accuracy sets explicit row-dependent parameters.

Lemma 2 (Output error gives score geometry). *Let $i \in [n-1]$, let $0 < \varepsilon < 1/(2i)$, and suppose $\max_j |A(i,j) - T(i,j)| \le \varepsilon$. Then row i satisfies (3) on every $I \subseteq [n-1]$ containing i, with*

$$\delta_i = \log \frac{1 + i\varepsilon}{1 - i\varepsilon}, \qquad \gamma_i = \log \frac{1 - i\varepsilon}{i\varepsilon}. \tag{4}$$

This is a log-odds conversion: near-uniform prefix probabilities give near-equal prefix scores, while the first future probability gives the boundary gap. Appendix A gives the proof.

The lower bound uses only entrywise matrix error, which follows from the operational row-ℓ_1 bound in (1); the upper construction attains the row-ℓ_1 target. Therefore $\|x\|_\infty \le \|x\|_q \le \|x\|_1$ gives the same two-term law for maximum row-ℓ_q error at every fixed $1 \le q \le \infty$.

The scale $1/n$ has a direct interpretation. For a binary value stream, $i(Tv)_i$ is an integer prefix sum. Rounding $i(Av)_i$ recovers that sum whenever the scalar output error is below $1/(2i)$. The eligibility condition $i\varepsilon < 1/2$ marks this recovery scale. Fixed targets n^{-p}, $p > 1$, give every nonterminal boundary ($i \le n-1$) a polynomial margin.

4 The rank–logit law

Lemma 2 converts output accuracy into row profiles. The following nonasymptotic determinant bound converts those profiles into a constraint on rank and logit radius, while Theorem 4 supplies the matching construction.

Theorem 3 (Row-subset determinant bound). *Let $\varnothing \ne I \subseteq [n-1]$, $m = |I|$, and suppose the rows in I satisfy (3) on I, with positive δ_i, γ_i. If $r \ge 1$ is an integer, $\rho(S) \le r$ and $B(S) \le B$, then*

$$\prod_{i \in I} \gamma_i \le \sum_{k=1}^{\min\{r,m\}} k! \left\{ {m \atop k} \right\} (2B)^k \max_{\substack{R \subseteq I \\ |R| = k}} \prod_{i \in I \setminus R} \delta_i. \tag{5}$$

If additionally $\delta_i \le \delta$ and $\gamma_i \ge \gamma$ on I, then

$$\gamma^m \le \sum_{k=1}^{\min\{r,m\}} k! \left\{ {m \atop k} \right\} \delta^{m-k} (2B)^k. \tag{6}$$

Take adjacent score differences. The selected matrix has low rank, a large diagonal, and a small strict lower triangle. The key move is to expand the determinant of its upper-triangular part, rather than the determinant of the low-rank matrix itself; Appendix B gives the proof.

The theorem can be applied to any subset of eligible rows, namely rows with $i\varepsilon < 1/2$. Our main result uses all $n-1$ nonterminal rows, which are eligible when $\varepsilon = o(1/n)$.

For $0 < \varepsilon < 1$ and an integer $1 \le r \le n-1$, define

$$B_r^{\mathrm{out}}(n, \varepsilon) = \inf \left\{ B(S) : \rho(S) \le r, \sup_{\|v\|_\infty \le 1} \|(A-T)v\|_\infty \le \varepsilon \right\}, \qquad B_{r,p}^{\mathrm{out}}(n) = B_r^{\mathrm{out}}(n, n^{-p}).$$

Write $N = n-1$ through the rest of this section.

Theorem 4 (Balanced rank-r construction)**.** *Let $n \ge 2$, $0 < \varepsilon \le 1/n$, $N = n-1$, and let r be an integer with $1 \le r \le N$. There is $S \in \mathbb{R}^{n\times n}$ with $\operatorname{rank}(S) = \rho(S) = r$, output error at most ε, and*

$$\log B(S) \le \frac{N}{r} \log \frac{64e \log(8n/\varepsilon)}{n\varepsilon} + \log\log \frac{8n}{\varepsilon}. \tag{7}$$

The construction partitions the nonterminal rows into r balanced consecutive blocks and uses one outer product per block. Appendix D verifies its rank, error, and radius.

Theorem 5 (Two-term fixed-rank output law)**.** *For every fixed $p > 1$ and fixed positive integer r,*

$$\log B_{r,p}^{\mathrm{out}}(n) = \frac{n-1}{r} \big((p-1)\log n + \log\log n + O_{p,r}(1) \big). \tag{8}$$

Proof. We first apply the determinant bound to every feasible score matrix and then use Theorem 4 for the matching construction.

Set $\varepsilon = n^{-p}$ and $N = n-1$. For the lower bound, fix any feasible S in the definition of $B_{r,p}^{\mathrm{out}}(n)$, put $B = B(S)$, and set

$$x_n = N\varepsilon, \qquad \delta_n = \log \frac{1+x_n}{1-x_n}, \qquad \gamma_n = \log \frac{1-x_n}{x_n}.$$

Since $p > 1$, $x_n \to 0$. For all sufficiently large n, operational error at most ε and Lemma 2 give, uniformly on $I = [N]$,

$$\delta_i \le \delta_n = 2n^{1-p}(1+o(1)), \qquad \gamma_i \ge \gamma_n = (p-1)\log n + o(1).$$

Eventually $\gamma_n \ge \delta_n$ and $N \ge r$. Since row centering preserves score differences, $S(i,i) - S(i,i+1) \ge \gamma_n$ in every profiled row implies $2B \ge \gamma_n$. Hence, for $k \le r$,

$$k! \left\{ {N \atop k} \right\} \le k^N \le r^N, \qquad \delta_n^{N-k}(2B)^k \le \delta_n^{N-r}(2B)^r.$$

Applying Theorem 3 and summing its at most r terms gives

$$\gamma_n^N \le r^{N+1} \delta_n^{N-r} (2B)^r,$$

and therefore

$$\log B \ge \frac{N}{r} \log \frac{\gamma_n}{r\delta_n} + \log \frac{\delta_n}{2} - \frac{1}{r}\log r.$$

Here

$$\log \frac{\gamma_n}{r\delta_n} = (p-1)\log n + \log\log n + O_{p,r}(1),$$

while, after normalization by N/r, the remaining two terms are $O_{p,r}(\log n/n) = o(1)$. Taking the infimum over all feasible S proves the lower estimate; no minimizer is needed.

For the upper estimate, Theorem 4 gives

$$\log B \leq \frac{N}{r} \log \frac{64eL_n}{n\varepsilon} + \log L_n, \qquad L_n = \log \frac{8n}{\varepsilon}.$$

At $\varepsilon = n^{-p}$,

$$\log \frac{64eL_n}{n\varepsilon} = (p-1)\log n + \log\log n + O_p(1),$$

and $(r/N)\log L_n = O_{p,r}(\log\log n/n) = o(1)$. Combining the two bounds proves (8). ☐

The $O_{p,r}(1)$ term permits a multiplicative $\exp(O(n/r))$ uncertainty in B.

Corollary 6 (Head-dimension tradeoff)**.** *Fix $p > 1$ and a positive integer d_h. A position-dependent unmasked pure dot-product head of dimension d_h with operational output error at most n^{-p} obeys, for some constant C_{p,d_h} and all sufficiently large n,*

$$d_h \log B(S) \geq (n-1)\big((p-1)\log n + \log\log n - C_{p,d_h}\big).$$

Proof. Writing $S = QK^\top$ gives $\rho(S) = \operatorname{rank}(SP_0) \leq \operatorname{rank}(S) \leq d_h$. Thus S is feasible in the definition of $B^{\text{out}}_{d_h,p}(n)$, and $B(S) \geq B^{\text{out}}_{d_h,p}(n)$. The lower bound in Theorem 5, applied with the fixed rank parameter d_h, gives the displayed inequality. ☐

Theorem 5 keeps r fixed. If r grows, the upper bound remains uniform for $r = o(N/\log\log n)$, while the determinant lower bound contains an additional $-\log r$ term. Thus the two diverging terms still match when $\log r = o(\log\log n)$. Appendix C gives a complementary estimate for larger ranks.

At full centered rank, Proposition 8 attains $\log B = \log\log(n/\varepsilon) + O(1)$. The closed-form bound (7) is loose at $r = N$: its singleton-block construction itself has $\log B = \log\log n + O_p(1)$.

Corollary 7 (Polynomial radius budget)**.** *Fix $p > 3/2$ and $c > 0$. A score matrix with $B(S) \leq n^c$ and output error at most n^{-p} has*

$$\rho(S) \geq \left(\frac{p - 3/2}{c + p - 1} - o(1)\right) n.$$

Appendix C replaces the rank-dependent permutation count by a columnwise Hadamard bound and proves the corollary uniformly over the rank. The resulting estimate yields a positive coefficient for $p > 3/2$. Behavior at the threshold remains open.

The determinant bound separates the accuracy regimes more precisely than a single asymptotic formula.

accuracy ε	eligible rows	status for fixed r
n^{-p}, fixed $p > 1$	all nonterminal	two-term law, Theorem 5
general $o(1/n)$	all nonterminal	two bounds with unmatched leading terms
c/n, fixed $c > 0$	linear subset	$\Omega_{c,r}(n/r)$ to $O_{c,r}(n\log\log n/r)$
$1/n \ll \varepsilon \ll 1$	$\Theta(1/\varepsilon)$	lower bound; construction open

At $\varepsilon = 1/(n\log n)$, the determinant bound gives $(N/r)(\log\log n + \log\log\log n + O(1))$. The budget in (7) gives $(N/r)(2\log\log n + O(1))$. Their leading coefficients differ, which is why (8) is restricted to polynomial targets. The behavior of B^{out}_r at and above the $1/n$ scale remains open.

Full centered rank changes the scaling.

Proposition 8 (Full-rank two-level scores)**.** *Let $n \geq 2$. For $\Gamma > 0$, set*

$$S_\Gamma(i,j) = -\Gamma\,\mathbf{1}[j > i].$$

Then $\operatorname{rank}(S_\Gamma P_0) = n-1$, $B(S_\Gamma) \leq \Gamma$, and every nonterminal row i has a causal $(0,\Gamma)$ profile on every $I \subseteq [n-1]$ containing i. Taking $\Gamma = \log(2(n-1)/\varepsilon)$, for $0 < \varepsilon \leq 1$, gives output error at most ε.

Proof. The submatrix on rows $1,\dots,n-1$ and columns $2,\dots,n$ is upper triangular with diagonal $-\Gamma$, so $\operatorname{rank}(S_\Gamma) \ge n-1$. Column 1 of S_Γ is zero, since $1 > i$ fails for every $i \ge 1$, so $e_1 \in \ker S_\Gamma$, the rank is $n-1$, and $\ker S_\Gamma = \operatorname{span}\{e_1\}$. If $S_\Gamma P_0 x = 0$, then $P_0 x$ lies in this kernel and has coordinate sum zero, which gives $P_0 x = 0$ and $\ker(S_\Gamma P_0) = \operatorname{span}\{\mathbf{1}\}$. The centered rank is $n-1$. Row centering keeps every entry within the raw row range Γ. If λ_i is the future attention mass in row i, then

$$\lambda_i = \frac{(n-i)e^{-\Gamma}}{i + (n-i)e^{-\Gamma}} \le (n-1)e^{-\Gamma} \le \varepsilon/2.$$

Conditioned on the prefix, the attention distribution is uniform. Its row-ℓ_1 error therefore equals $2\lambda_i \le \varepsilon$. $\square$

The matrix S_Γ gives a finite-Γ approximation to the causal mask. It uses full centered rank and a growing realized radius to approximate the architectural support boundary.

5 Determinant mechanism and matching construction

Let $N = n-1$, let $\Delta \in \mathbb{R}^{n\times N}$ be the first-difference operator, and take the first N rows of $S\Delta$:

$$D(i,j) = S(i,j+1) - S(i,j).$$

Since $P_0\Delta = \Delta$, $\operatorname{rank}(D) \le \rho(S)$. Row i's profile gives a small strict lower triangle and a large diagonal:

$$|D(i,j)| \le \delta_i \quad (j < i), \qquad D(i,i) \le -\gamma_i.$$

A direct determinant argument now fails: when $\operatorname{rank}(D) < N$, it only says $\det D = 0$. Instead let U be the upper-triangular part of D and write $U = D - E$, where E is strictly lower triangular. The determinant of U is large because every diagonal entry $U(i,i) = D(i,i)$ has magnitude at least γ_i. In its column expansion, rank allows at most r columns from D; every other column comes from the small lower triangle. Lemma 12 shows that $k!\left\{{N \atop k}\right\}$ pairs (π, K) can contribute when k columns are taken from D. Keeping the associated row indices yields (5); Appendix B supplies the complete argument, including arbitrary row subsets.

The construction balances row budgets over r consecutive blocks. One outer product creates the score drops inside each block. Appendix D gives the factorization and verifies its centered rank, row error, and radius bound (7).

6 Numerical checks and query/key coordinate bounds

The audit evaluates the explicit construction and the determinant bound at $p = 2$. High-precision decimal arithmetic computes the row-ℓ_1 error and centered radius. The lower curve solves (5) with the row profile from Lemma 2 and the rank-independent bound.

Every quantity in the bound is measurable on a realized score matrix: per-row oscillations and drops, the centered radius, and the centered rank can be read off logits directly, so (5) evaluates at any (n, r, ε) with no asymptotic step. It gives a necessary, not sufficient, condition for accuracy ε at the measured resources.

For a radius value B, define its normalized remainder at $p = 2$, the value used throughout this section, by

$$R_{n,r}(B) = \frac{r}{n-1}\log B - \big(\log n + \log\log n\big),$$

the general leading term being $(p-1)\log n + \log\log n$. Theorem 5 predicts $R_{n,r} = O_r(1)$ for fixed r. Figure 1 places the construction between the lower bound and its closed-form budget for all 36 tested parameter pairs. Every measured row error is below $0.251\,n^{-2}$. The displayed interval bounds the unknown optimum.

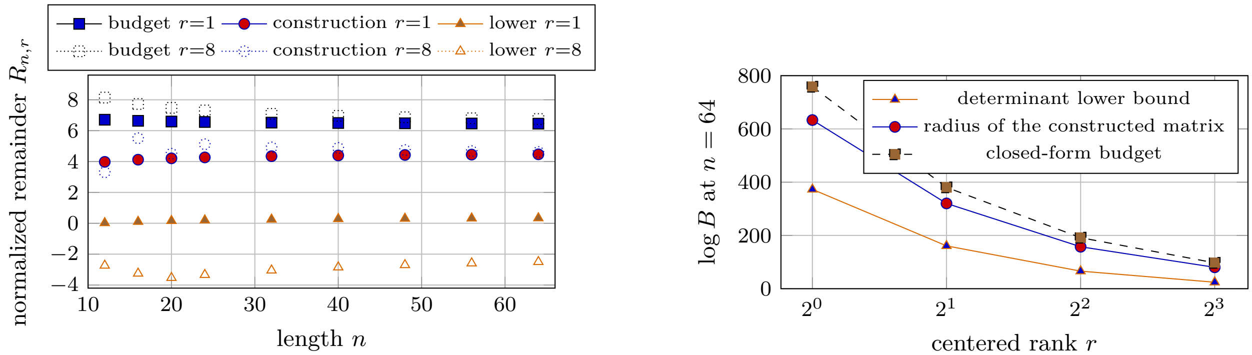

Figure 1: Finite-size construction and bracket at $p = 2$. Left: the three normalized remainders as n varies, with $r = 1$ solid and $r = 8$ dotted. At $n = 64$, the full bracket spans 6.12 nats for $r = 1$ and 9.25 for $r = 8$. Right: the row-weighted determinant lower bound, radius of the constructed matrix, and closed-form budget at $n = 64$.

6.1 Consequences of bounded query/key coordinates

Proposition 1 converts the determinant lower bound into a statement about heads whose query and key coordinates are bounded.

Corollary 9 (Infeasibility under bounded query/key coordinates)**.** *Fix $p > 1$, a positive integer d_h, and a floating-point format whose largest finite magnitude is $F > 0$. Let $Q, K \in \mathbb{R}^{n\times d_h}$ have entries bounded in magnitude by F, and set $S = QK^\top$. Suppose the operational output error is at most $\varepsilon = n^{-p}$. For every nonempty $I \subseteq [n-1]$ on which $i\varepsilon < 1/2$, set $m = |I|$ and*

$$\delta_i = \log \frac{1 + i\varepsilon}{1 - i\varepsilon}, \qquad \gamma_i = \log \frac{1 - i\varepsilon}{i\varepsilon}.$$

Then necessarily

$$\prod_{i\in I} \gamma_i \le \sum_{k=1}^{\min\{d_h,m\}} k! \left\{ {m \atop k} \right\} (4d_h F^2)^k \max_{\substack{R\subseteq I \\ |R|=k}} \prod_{i\in I\setminus R} \delta_i.$$

Proof. Operational error implies entrywise error at most ε, so Lemma 2 supplies the displayed profiles. Moreover $\rho(S) \le \operatorname{rank}(S) \le d_h$, while Proposition 1 gives $B(S) \le 2d_h F^2$. Substitution in Theorem 3 proves the claim. □

Here the floating-point format is used only to set the coordinate bound F; the matrix product and softmax are evaluated in exact arithmetic. Evaluation of the test at $p = 2$ gives the table. Each entry is the largest n not excluded by the bound. The bfloat16 and float32 thresholds are computed from their respective maxima ($\log F = 88.7189$ and 88.7228); both round to the displayed value, and the nonexcluded lengths coincide.

format	$\log F$	$d_h = 8$	32	64	128
float16	11.09	65	262	531	1075
bfloat16	88.72	268	1053	2093	4165
float32	88.72	268	1053	2093	4165

The first failing length remains excluded as n grows because each δ_i decreases and each γ_i increases on the same eligible row subset. At $n = 8192$, the bound gives $\log B \ge 418.72$ for $r = 128$. Proposition 1 then forces

$$\max_i \|q_i\|_2 \max_j \|k_j - \bar{k}\|_2 \ge e^{418.72}.$$

The same calculation excludes head dimensions through 252 for float32 or bfloat16 and through 934 for float16.

For scale, deployed decoder contexts run from 1.3×10^5 to beyond 10^6 positions (Llama Team, AI @ Meta, 2024; Gemini Team, Google, 2024); every nonexcluded length in the table lies below 4.2×10^3, and the reference length $n = 8192$ is a sixteenth of the shorter figure.

7 Boolean query/key adaptation

The bounds so far hold one score matrix fixed while value streams vary. In a transformer, changing the input also changes the queries and keys, and hence the attention matrix. We ask whether changing this input can avoid the costs imposed by our main theorem. The answer is no in the following Boolean model.

Position j carries a payload bit $v_j \in \{0,1\}$ and the embedding $x_j = e_j + v_j u_j$, with free position-indexed vectors $e_j, u_j \in \mathbb{R}^d$. A head computes $q_i = Qx_i$, $k_j = Kx_j$ with $Q, K \in \mathbb{R}^{d_h \times d}$, applies the unmasked row softmax $A(v)$ of $S(v)_{ij} = \langle q_i, k_j \rangle$, and outputs $(A(v)v)_i$; the value stream is the payload itself. Each score $S(v)_{ij}$ depends only on v_i and v_j. In particular, when $v_i = 0$, changing v_j for $j \neq i$ changes only column j of row i. The head is *ε-accurate* when $|(A(v)v)_i - \frac{1}{i}\sum_{j \le i} v_j| \le \varepsilon$ for every payload $v \in \{0,1\}^n$ and every nonterminal row i. Write $B_\infty = \max_v B(S(v))$; in particular $B(S(0)) \le B_\infty$. Every realized score matrix factors through the head, so $\rho(S(v)) \le d_h$ for every v.

Theorem 10 (Adaptive extraction)**.** *There are absolute constants $i_0 \in \mathbb{N}$ and $c_0, C > 0$ with the following property. If a head is ε-accurate, then every row $i_0 \le i \le n-1$ with $i\varepsilon \le c_0$ satisfies*

$$\max_j |A(0)_{ij} - T(i,j)| \le C\varepsilon.$$

The difficulty is that activating v_j changes both its value and its key, so a Boolean input does not directly reveal $A(0)_{ij}$. We restrict to $v_i = 0$, which fixes the query in row i. At each of two different prefix counts, we compare Boolean inputs having the same number of active prefix bits. Together, these comparisons separate the contributions of the baseline and activated scores. Swapping one active prefix bit shows that the baseline prefix weights are nearly equal; summing over cyclic windows determines their total; and activating all future bits bounds the future mass. Lemma 2 then converts the resulting entrywise estimate into the score profile required by Theorem 3. Appendix E gives the proof with explicit constants.

For $0 < \varepsilon < 1$ and a positive integer r, let $B_r^{\mathrm{ad}}(n,\varepsilon)$ denote the infimum of B_∞ over all ε-accurate heads with $d_h \le r$, allowing d and all e_j, u_j, Q, K to vary. Set $B_{r,p}^{\mathrm{ad}}(n) = B_r^{\mathrm{ad}}(n, n^{-p})$.

Corollary 11 (Fixed-rank law under Boolean adaptation)**.** *For every fixed $p > 1$ and fixed positive integer r,*

$$\log B_{r,p}^{\mathrm{ad}}(n) = \frac{n-1}{r}\big((p-1)\log n + \log\log n + O_{p,r}(1)\big).$$

The lower bound applies Theorem 3 to the all-zero score matrix $S(0)$, whose centered rank is at most r; uniform accuracy over Boolean inputs is used only to force the required profile on $S(0)$. The upper bound embeds the fixed construction of Theorem 4 with $u_j = 0$. For this model, adaptation leaves both diverging terms of the fixed-rank law unchanged. Appendix E gives the proof.

8 Open problems

The fixed-rank law leaves three main questions. First, what happens when rank grows with sequence length? Resolving this would also determine whether polynomially bounded logits force rank proportional to n throughout $p > 1$, rather than only for $p > 3/2$. Second, what is the correct logit cost when the error is of order $1/n$, where our lower and upper bounds do not match? Third, can rank be replaced by a noise-stable notion of effective rank, so that the bound can be applied to score matrices measured in trained models?

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A Output error and score geometry

Proof of Lemma 2. For $j \le i$, $T(i,j) = 1/i$, so entrywise error at most ε gives

$$\frac{1 - i\varepsilon}{i} \le A(i,j) \le \frac{1 + i\varepsilon}{i}.$$

Score differences are logarithms of probability ratios. In particular, for $j, t \le i$,

$$S(i,j) - S(i,t) = \log \frac{A(i,j)}{A(i,t)}, \qquad |S(i,j) - S(i,t)| \le \log \frac{1 + i\varepsilon}{1 - i\varepsilon}.$$

This bounds the oscillation on every subset of prefix columns by δ_i. For the adjacent key, $T(i, i+1) = 0$ gives $A(i, i+1) \le \varepsilon$, while $A(i,i) \ge (1 - i\varepsilon)/i$, which yields

$$S(i, i+1) - S(i,i) = \log \frac{A(i, i+1)}{A(i,i)} \le -\log \frac{1 - i\varepsilon}{i\varepsilon}.$$

□

B Determinant proof

Lemma 12 (Permitted permutation count)**.** *For $1 \le k \le m$,*

$$\#\big\{(\pi, K) : K \subseteq [m],\ |K| = k,\ \pi \in \mathfrak{S}_m,\quad \pi(j) > j \text{ for all } j \notin K\big\} = k!\left\{ {m \atop k} \right\}.$$

Proof. We construct a bijection between the counted pairs (π, K) and pairs consisting of a partition of $[m]$ into k blocks and a bijection from the block maxima to the block minima.

The largest element j of any cycle of π has $\pi(j) \le j$ and hence lies in K, so iterating π from any point meets K. Let $f(x)$ be the first element of K on the orbit $x, \pi(x), \pi^2(x), \ldots$; thus $f(x) = x$ on K. The fibers of f partition $[m]$ into k blocks, and the block $F_a = f^{-1}(a)$ meets K in $\{a\}$.

Fix $a \in K$ and write $F_a = \{y_1 < \cdots < y_t\}$. Each $x \in F_a \setminus \{a\}$ lies outside K, so $\pi(x) > x$; and $\pi(x) \in F_a$, since either $\pi(x) = a$ or $\pi(x) \notin K$ and its orbit first meets K at a. So π maps $F_a \setminus \{a\}$ injectively into F_a with $\pi(x) > x$. The largest element y_t admits no such image, so $a = y_t = \max F_a$. The domain $F_a \setminus \{a\}$ and the set $\{y_2, \ldots, y_t\}$ both have $t-1$ elements, and $\pi(x) > x$ prevents y_1 from being an image. Thus π maps the former set bijectively onto the latter. If $t = 1$, there is nothing more to show. If $t \ge 2$, then $\pi(y_{t-1}) = y_t$. Descending induction gives the entire chain: once $y_{j+2}, \ldots, y_t$ are the images of $y_{j+1}, \ldots, y_{t-1}$, the only unused image larger than y_j is y_{j+1}. Hence $\pi(y_j) = y_{j+1}$. Thus off K the permutation is the successor map of the blocks, its images are the nonminimal elements of the blocks, and π must map K, the set of block maxima, bijectively onto the set of block minima.

Conversely, let Π be a partition of $[m]$ into k blocks and σ a bijection from the block maxima onto the block minima. Take K to be the set of block maxima and let π act as the successor map within blocks on $[m] \setminus K$ and as σ on K. Every nonminimal element has one preimage, its block predecessor, and every block minimum has one under σ; so $\pi \in \mathfrak{S}_m$, and $\pi(j) > j$ off K. From any nonmaximal element the successor chain climbs within its block and meets K first at the block maximum, so the fibers of f recover Π, and the restriction of π to K recovers σ. The two constructions are therefore mutually inverse, and the pairs (π, K) are in bijection with the pairs (Π, σ). Since $\left\{ {m \atop k} \right\}$ counts the partitions of $[m]$ into k nonempty blocks, there are $k!\left\{ {m \atop k} \right\}$ such pairs. ☐

Proof of Theorem 3. Let $N = n - 1$, let $\Delta \in \mathbb{R}^{n \times N}$ be the first-difference operator, and set

$$D = (SP_0)\Delta = S\Delta, \qquad D(i,j) = S(i, j+1) - S(i,j).$$

Each column of Δ sums to zero, so $P_0\Delta = \Delta$; the representation $D = (SP_0)\Delta$ gives the rank and entry bounds, while $D = S\Delta$ gives the formula for $D(i,j)$. Then $\operatorname{rank}(D) \le \rho(S)$, and every entry of D has magnitude at most $2B$. Order $I = \{i_1 < \cdots < i_m\}$ and define $D_I = (D(i_a, i_b))_{a,b=1}^m$. This submatrix has rank at most r. For $b < a$, both i_b and $i_b + 1$ lie in I^+ and are at most i_a; recall that the profile bounds the oscillation of $S(i,j)$ over $j \in I^+$, $j \le i$, by δ_i and gives $S(i, i+1) \le S(i,i) - \gamma_i$, so the profile of row i_a yields

$$|D_I(a,b)| \le \delta_{i_a}, \qquad D_I(a,a) \le -\gamma_{i_a}.$$

Let U_I be the upper-triangular part of D_I, and let $E_I = D_I - U_I$. Thus E_I is strictly lower triangular and

$$|\det U_I| \ge \prod_{i \in I} \gamma_i, \qquad U_I = D_I - E_I.$$

Expand $\det U_I = \det(D_I - E_I)$ multilinearly in its m columns. For $K \subseteq [m]$, take column j from D_I when $j \in K$ and from E_I otherwise; write M_K for the resulting matrix. Signs will be discarded after taking absolute values. If $|K| > r$, the selected D_I-columns are linearly dependent, so the mixed determinant vanishes. If $K = \varnothing$, the mixed matrix is strictly lower triangular and its determinant also vanishes.

Fix $|K| = k$. In the Leibniz expansion, a product can be nonzero only when $\pi(j) > j$ for every $j \notin K$, because column j of E_I is supported strictly below row j. Lemma 12 counts the pairs (π, K) with this property.

Every such product uses k entries from D_I, each bounded by $2B$. An entry from E_I in row a is bounded by δ_{i_a}. A determinant product uses each row once, so if its D_I-entries occupy the original rows in a k-element set $R \subseteq I$, its E_I-entries contribute at most

$$\max_{\substack{R\subseteq I\\ |R|=k}} \prod_{i\in I\setminus R} \delta_i.$$

Summing absolute values proves (5); replacing all δ_i, γ_i by their common bounds proves (6). $\square$

C Hadamard bound and rank-independent radius lower bound

Proposition 13 (Hadamard bound and rank-independent radius lower bound)**.** *Under the row-dependent hypotheses of Theorem 3,*

$$B \geq \frac{1}{2}\max_{i\in I} \gamma_i.$$

In particular, under the constant-profile hypotheses, $B \geq \gamma/2$. If, in addition, $m = |I|$, $\delta \leq \gamma$, and $r \leq m$, then

$$\gamma^m \leq 2^{m-1}(\delta\sqrt{m})^{m-r}(2B\sqrt{m})^r.$$

Proof. For the radius bounds, row centering preserves score differences, so every profiled row satisfies

$$\gamma_i \leq |(SP_0)(i,i) - (SP_0)(i,i+1)| \leq 2B(S) \leq 2B.$$

Maximizing over $i \in I$ proves the general statement and its constant-profile specialization.

For the determinant bound, assume $\delta \leq \gamma$ and $r \leq m$, so that the radius bound gives $2B \geq \gamma \geq \delta$. Use the column expansion from the proof of Theorem 3. As there, $\gamma^m \leq |\det U_I| \leq \sum_K |\det M_K|$, the sum over column subsets K drawn from D_I. Column m of E_I is zero, so every term taking column m from E_I vanishes; terms with $|K| > r$ vanish by rank. At most 2^{m-1} terms remain, one for each set $K \ni m$ with $|K| \leq r$. In the ordered m-by-m submatrix, every column of E_I has at most $m-1$ possibly nonzero entries, each bounded by δ, hence norm at most $\delta\sqrt{m}$; every column of D_I has norm at most $2B\sqrt{m}$. For a term using the columns K from D_I, with $|K| = k$, Hadamard's inequality gives

$$|\det M_K| \leq (\delta\sqrt{m})^{m-k}(2B\sqrt{m})^k \leq (\delta\sqrt{m})^{m-r}(2B\sqrt{m})^r,$$

the second step because $k \leq r$ and $\delta \leq 2B$. Since $|\det U_I| \geq \prod_{i\in I}\gamma_i \geq \gamma^m$, summing the remaining terms proves the determinant bound. $\square$

Proof of Corollary 7. Set $N = n-1$, $\varepsilon = n^{-p}$, $r = \rho(S)$, and $B = B(S)$. Operational error implies entrywise error at most ε. In the row-dependent profile supplied by Lemma 2, δ_i increases and γ_i decreases with i. Thus, for all sufficiently large n, the values at $i = N$ give the following common profile on $I = [N]$:

$$\delta = \log\frac{1+N\varepsilon}{1-N\varepsilon}, \qquad \gamma = \log\frac{1-N\varepsilon}{N\varepsilon}.$$

Here $\gamma > 0$ forces $r \geq 1$, and $r \leq N$ holds unconditionally; eventually $\gamma \geq \delta$. Proposition 13 then gives $2B \geq \gamma \geq \delta$ and

$$\gamma^N \leq 2^{N-1}(\delta\sqrt{N})^{N-r}(2B\sqrt{N})^r.$$

Taking logarithms and isolating the rank-dependent term gives

$$r\log\frac{2B}{\delta} \geq N\log\frac{\gamma}{\delta\sqrt{N}} - (N-1)\log 2.$$

The profile parameters satisfy

$$\delta = 2n^{1-p}(1+o(1)), \qquad \gamma = (p-1)\log n + O_p(1), \qquad \sqrt{N} = n^{1/2}(1+o(1)).$$

Therefore

$$\log \frac{\gamma}{\delta\sqrt{N}} = (p - 3/2)\log n + \log\log n + O_p(1).$$

On the other hand, $B \le n^c$ and $1/\delta = \frac{1}{2}n^{p-1}(1 + o(1))$ give

$$0 \le \log \frac{2B}{\delta} \le (c + p - 1)\log n + O_p(1).$$

Replacing the nonnegative multiplier of r by this upper bound can only weaken the rank lower bound. Dividing the resulting inequality by $N \log n$, and using $N/n \to 1$, proves

$$r \ge \left(\frac{p - 3/2}{c + p - 1} - o(1)\right) n.$$

All remainders depend only on the fixed p, c, so this estimate is uniform in the possibly n-dependent rank r. $\square$

D The upper construction

Proof of Theorem 4. We choose a small allowable variation δ_i among the prefix scores and a gap γ_i separating the future scores. One outer product will handle each block of rows, and the partition will balance the largest radius produced by these factors.

For $1 \le i \le N$, set

$$\delta_i = \frac{i\varepsilon}{16}, \qquad \gamma_i = \log \frac{8(n-i)}{i\varepsilon}, \qquad w_i = \log\left(1 + \frac{\gamma_i}{\delta_i}\right),$$

and write $L_\varepsilon = \log(8n/\varepsilon)$. Using $i\varepsilon \le N/n < 1 \le n - i$ and $n - i \le in$, every row satisfies the following bounds, which will be used repeatedly:

$$\delta_i \le \frac{1}{16} < \log 8 \le \gamma_i \le L_\varepsilon, \qquad (n-i)e^{-\gamma_i} = \frac{i\varepsilon}{8}. \tag{9}$$

Empty sums are zero.

Balanced partition. The row budget grows multiplicatively by $1 + \gamma_t/\delta_t$, so equal-length blocks can have very different radii. We instead balance their logarithmic costs w_t, which controls the largest block product. Partition $[N]$ into r consecutive nonempty blocks so that, for every $I = \{a, \ldots, b\}$,

$$\sum_{t=a+1}^{b} w_t \le W/r, \qquad W = \sum_{i=1}^{N} w_i.$$

A left-to-right greedy scan through $w_2, \ldots, w_N$ starts a new block before t whenever including w_t would make the current block's internal load exceed W/r. At such a cut, the completed internal load together with w_t exceeds W/r. The weight w_t becomes the first index of the new block and is not counted in that block's internal load, so the groups associated with different cuts are disjoint. Their total weight is at most W, and therefore there are fewer than r cuts. If the resulting partition has fewer than r blocks, $r \le N$ guarantees a nonsingleton block. Splitting such blocks reaches r and only removes internal weights.

Construction. For one block, v^I accumulates the required score drops and u^I rescales each row. Define

$$c_a = 1, \qquad G_t = \sum_{s=a}^{t} \gamma_s c_s, \qquad c_t = \frac{G_{t-1}}{\delta_t} \quad (t > a),$$

$$u_i^I = \frac{\mathbf{1}[i \in I]}{c_i}, \qquad v_j^I = -\sum_{t=a}^{\min\{b, j-1\}} \gamma_t c_t, \qquad S = \sum_I u^I (v^I)^\top.$$

The definitions form a joint recursion in increasing t: $G_a = \gamma_a$, and $G_t = G_{t-1}(1 + \gamma_t/\delta_t)$ for $t > a$. By (9), consecutive G's grow by the factor $1 + \gamma_t/\delta_t \geq 2$, and $c_i \geq 1$ because $\delta_i \leq 1/16 < \log 8 \leq G_a \leq G_{i-1}$ for $i > a$. For $i \in I$, row i of S is v^I/c_i:

$$S(i,j) = \begin{cases} 0, & j \leq a, \\ -G_{j-1}/c_i, & a < j \leq b+1, \\ -G_b/c_i, & j > b+1. \end{cases}$$

If $i = a$, all prefix scores $S(i,j)$, $j \leq i$, are zero. If $i > a$, they decrease from 0 to $S(i,i) = -G_{i-1}/c_i = -\delta_i$. In either case, every future score $S(i,j)$, $j > i$, is at most $-G_i/c_i = S(i,i) - \gamma_i \leq -\gamma_i$. Row n of S is zero.

Rank. S is a sum of r outer products, so $\rho(S) \leq \operatorname{rank}(S) \leq r$. Suppose $\sum_I a_I P_0 v^I = 0$. Then $\sum_I a_I v^I$ lies in $\ker P_0$, so it is constant. Fix a block I with first row a. Across coordinates a and $a+1$, every v^J with $J \neq I$ is unchanged, while $v^I_{a+1} - v^I_a = -\gamma_a c_a$. Constancy therefore gives

$$0 = \sum_J a_J (v^J_{a+1} - v^J_a) = -a_I \gamma_a c_a,$$

and $\gamma_a c_a > 0$ forces $a_I = 0$. Thus the vectors $P_0 v^I$ are linearly independent. At the first row a of block I, row a of S is $(v^I)^\top$, because $c_a = 1$. Hence the corresponding row of SP_0 is $(P_0 v^I)^\top$, so $\rho(S) \geq r$. Therefore $\operatorname{rank}(S) = \rho(S) = r$.

Output error. Fix $i \in I$, let $y_j = e^{S(i,j)} \leq 1$, and split the softmax mass of row i into $Z' = \sum_{j \leq i} y_j$ and $F = \sum_{j>i} y_j$, so $A(i,j) = y_j/Z$ with $Z = Z' + F$. The bound on the future scores and the identity $(n-i)e^{-\gamma_i} = i\varepsilon/8$ give $F \leq i\varepsilon/8$, while $1 - e^{-x} \leq x$ and the geometric growth of G bound the prefix deficit. If $i = a$, then $i - Z' = 0$. If $i > a$, then

$$i - Z' = \sum_{j \leq i} \big(1 - e^{S(i,j)}\big) \leq \sum_{j \leq i} |S(i,j)| = \frac{1}{c_i} \sum_{t=a}^{i-1} G_t \leq \frac{2G_{i-1}}{c_i} = 2\delta_i = \frac{i\varepsilon}{8},$$

because each G_t is at most half of G_{t+1}. Thus $i - Z' \leq i\varepsilon/8$ in every row. Rows $A(i,:)$ and $T(i,:)$ are probability vectors, so the error is twice the total undershoot. Write $x_+ = \max\{x, 0\}$. Since $T(i,j) = 0$ for $j > i$, only prefix indices contribute to the positive part:

$$\|A(i,:) - T(i,:)\|_1 = 2 \sum_j \big(T(i,j) - A(i,j)\big)_+ = 2 \sum_{j \leq i} \Big(\frac{1}{i} - \frac{y_j}{Z}\Big)_+.$$

Put $\bar{Z} = i(1 + \varepsilon/8)$. Since $Z' \leq i$ and $F \leq i\varepsilon/8$, we have $Z \leq \bar{Z}$, so each undershoot is at most $1/i - y_j/\bar{Z}$, which is nonnegative because $y_j \leq 1 \leq \bar{Z}/i$. Summing over the prefix,

$$\|A(i,:) - T(i,:)\|_1 \leq 2\Big(1 - \frac{Z'}{\bar{Z}}\Big) = \frac{2\big((i - Z') + i\varepsilon/8\big)}{\bar{Z}} \leq \frac{2}{i} \cdot \frac{i\varepsilon}{4} = \frac{\varepsilon}{2}.$$

Row n is zero in S, so its softmax is row n of T, and the worst-row error is at most $\varepsilon/2$.

Radius bound. Unrolling the recursion across a block and using $\gamma_a \leq L_\varepsilon$ with the balanced loads,

$$\log G_b = \log \gamma_a + \sum_{t=a+1}^{b} w_t \leq \log L_\varepsilon + \frac{W}{r}.$$

Since $\gamma_t/\delta_t \geq 1$ by (9), each $w_t \leq \log(2\gamma_t/\delta_t) \leq \log \frac{32L_\varepsilon}{t\varepsilon}$, so $N! \geq (N/e)^N$ and $n \leq 2N$ give

$$W \leq N \log \frac{32L_\varepsilon}{\varepsilon} - \log N! \leq N \log \frac{32eL_\varepsilon}{N\varepsilon} \leq N \log \frac{64eL_\varepsilon}{n\varepsilon}.$$

Every $c_i \geq 1$, so all entries of a block row v^I_j/c_i lie in $[-G_b, 0]$; subtracting the row mean leaves every magnitude at most the row range G_b, and row n is zero. Hence $B(S) \leq \max_I G_b$; the block bound on $\log G_b$ and the weight bound on W give (7). ☐

E Adaptive extraction

This appendix proves Theorem 10 and Corollary 11, and transfers the polynomial radius budget of Corollary 7 to the adaptive model (Corollary 14). The proof gives $i_0 = 12$, $c_0 = 10^{-3}$, and $C = 74$ in Theorem 10; these constants are not optimized.

Proof of Theorem 10. We first express every Boolean input using the scores from the all-zero input and the scores obtained after activating individual positions. Comparisons at two prefix counts then show that the baseline prefix probabilities are close to $1/i$, while a final comparison bounds the probability assigned to future positions.

Take $i_0 = 12$ and $c_0 = 10^{-3}$. Fix a row i with $12 \le i \le n-1$ and use only payloads $v = \mathbf{1}_X$ with $X \subseteq [n] \setminus \{i\}$. Every such payload has $v_i = 0$, so the query $q_i = Qe_i$ remains fixed, while key j takes two possible values: the baseline Ke_j when $v_j = 0$ and the switched $K(e_j + u_j)$ when $v_j = 1$. Thus row i has two possible scores in each column, selected by v_j. Using the normalizing constant from the all-zero input, set

$$Z = \sum_t e^{S(0)_{it}}, \qquad p_j = \frac{e^{S(0)_{ij}}}{Z}, \qquad \tilde{p}_j = \frac{e^{S(\mathbf{1}_{\{j\}})_{ij}}}{Z} \quad (j \ne i),$$

put $P_X = \sum_{j \in X} p_j$ and $\widetilde{P}_X = \sum_{j \in X} \tilde{p}_j$. Here $p_j = A(0)_{ij}$, while $\tilde{p}_j$ uses the score obtained when $v_j = 1$ but retains the normalizing constant Z from the all-zero input. Thus $\tilde{p}_j$ need not be a probability. Every p_j and $\tilde{p}_j$ is positive, and $1 - P_X = \sum_{t \notin X} p_t \ge p_i > 0$ for every admissible X. On the payload $\mathbf{1}_X$, the exponentiated scores sum to $Z\widetilde{P}_X$ over X and to $Z(1 - P_X)$ outside X. Since the values equal one precisely on X, the row-i output is

$$F(X) = \frac{\widetilde{P}_X}{(1 - P_X) + \widetilde{P}_X}. \tag{10}$$

Accuracy states $|F(X) - k/i| \le \varepsilon$ with $k = |X \cap [i-1]|$, since $v_i = 0$ removes bit i from the prefix sum. Only the prefix count of X enters the target; X may contain future positions.

A linear constraint at fixed prefix count. Write $Z_X = (1 - P_X) + \widetilde{P}_X$ for the denominator of (10) and suppose $k \le 2i/3$. Multiplying the accuracy bound by $Z_X > 0$ gives $\left|\widetilde{P}_X - \frac{k}{i} Z_X\right| \le \varepsilon Z_X$, and expanding Z_X rearranges the left side:

$$\widetilde{P}_X - \tfrac{k}{i}\big((1 - P_X) + \widetilde{P}_X\big) = \tfrac{i-k}{i}\Big(\widetilde{P}_X + h_k P_X - h_k\Big), \qquad h_k = \frac{k}{i-k}.$$

It remains to bound Z_X. Since $i\varepsilon \le c_0 \le 1$ and $i \ge 12$ force $\varepsilon \le \frac{1}{12}$, the output obeys $F(X) \le k/i + \varepsilon \le \frac{2}{3} + \frac{1}{12} = \frac{3}{4}$; then $\widetilde{P}_X = F(X) Z_X \le \frac{3}{4} Z_X$, so $Z_X = (1 - P_X) + \widetilde{P}_X \le 1 + \frac{3}{4} Z_X$, i.e. $Z_X \le 4$. With $\frac{i}{i-k} \le 3$ this yields the estimate used below:

$$\left|\widetilde{P}_X + h_k P_X - h_k\right| \le 12\,\varepsilon \qquad \text{whenever } k = |X \cap [i-1]| \le \tfrac{2i}{3}. \tag{11}$$

Why two prefix counts are needed. A Boolean input constrains the two sums on its support only through the single combination $\widetilde{P}_X + h_k P_X$ in (11). At one fixed count, a change in P_X can therefore be offset by a change in $\widetilde{P}_X$. In particular, the singleton Boolean input $F(\{j\})$ does not determine the baseline mass p_j. Using two counts changes the coefficient h_k and separates the two terms. Fix

$$k_1 = \lfloor i/3 \rfloor, \qquad k_2 = \lfloor 2i/3 \rfloor;$$

then $3k_1 \in \{i-2, i-1, i\}$ and $3k_2 \in \{2i-2, 2i-1, 2i\}$, and for $i \ge 12$ clearing denominators gives

$$h_{k_1} \le \tfrac{1}{2}, \qquad h_{k_2} \ge \tfrac{3}{2}, \qquad \frac{i-1}{k_1} \le \frac{10}{3}, \qquad \frac{i-1}{k_2} \le \frac{3}{2}, \tag{12}$$

The inequalities are verified at the end of the proof. In particular, $h_{k_2} - h_{k_1} \ge 1$. The swap step below also uses $k_2 \le i-2$, which likewise holds for $i \ge 12$. Throughout, $J = [i-1]$ and $L = \{i+1, \ldots, n\}$.

Comparing Boolean inputs with equal prefix counts. Two Boolean inputs with the same prefix count share the right side of (11), so subtracting their lines bounds the difference of the corresponding linear combinations by 24ε. We first compare prefix swaps and then compare inputs that differ on every future position.

Let $j \neq j' \in J$, let $k \in \{k_1, k_2\}$, and pick any $Y \subseteq J \setminus \{j, j'\}$ with $|Y| = k - 1$, possible since $k - 1 \leq i - 3$. The Boolean inputs $Y \cup \{j\}$ and $Y \cup \{j'\}$ cancel everything they share:

$$\big|(\tilde{p}_j - \tilde{p}_{j'}) + h_k\,(p_j - p_{j'})\big| \leq 24\varepsilon.$$

Subtracting the k_1 and k_2 versions (with possibly different Y for each count) removes $\tilde{p}_j - \tilde{p}_{j'}$, and dividing by $h_{k_2} - h_{k_1} \geq 1$ leaves

$$\operatorname{osc}_{j \in J}\, p_j \leq 48\varepsilon. \tag{13}$$

Next let $Y \subseteq J$ with $|Y| = k_2$. The Boolean inputs Y and $Y \cup L$ also share their count, so $|\widetilde{P}_L + h_{k_2} P_L| \leq 24\varepsilon$. Both terms are nonnegative, and $h_{k_2} \geq \frac{3}{2}$, so

$$P_L \leq 16\varepsilon. \tag{14}$$

Recovering the total prefix mass. Arrange J on a circle and let $W_1, \ldots, W_{i-1}$ be its cyclic windows of length $k \in \{k_1, k_2\}$; each $j \in J$ lies in k of them. Summing (11) over the windows and dividing by k gives, with $h_k/k = 1/(i-k)$,

$$\Big|\widetilde{P}_J + h_k P_J - \frac{i-1}{i-k}\Big| \leq \frac{i-1}{k}\,12\varepsilon,$$

at most 40ε for k_1 and 18ε for k_2 by (12). Subtracting the two displays cancels $\widetilde{P}_J$. The remaining constant terms satisfy

$$\frac{1}{i-k_2} - \frac{1}{i-k_1} = \frac{k_2 - k_1}{(i-k_1)(i-k_2)} = \frac{h_{k_2} - h_{k_1}}{i},$$

so $\big|(h_{k_2} - h_{k_1})\big(P_J - \frac{i-1}{i}\big)\big| \leq 58\varepsilon$; dividing by $h_{k_2} - h_{k_1} \geq 1$,

$$\Big|P_J - \frac{i-1}{i}\Big| \leq 58\varepsilon. \tag{15}$$

Completing the extraction. The mean of the prefix masses is $P_J/(i-1)$, which by (15) is within $58\varepsilon/(i-1) \leq 6\varepsilon$ of $1/i$, and each p_j is within the oscillation (13) of that mean, so

$$\big|p_j - \tfrac{1}{i}\big| \leq 48\varepsilon + 6\varepsilon = 54\varepsilon \qquad (j \in J).$$

Normalization gives $p_i = 1 - P_J - P_L$, so (15) and (14) imply $|p_i - \frac{1}{i}| \leq 58\varepsilon + 16\varepsilon = 74\varepsilon$. For $j \in L$, $0 < p_j \leq P_L \leq 16\varepsilon$, while $T(i,j) = 0$. Altogether

$$\max_j \big|A(0)_{ij} - T(i,j)\big| \leq C\varepsilon, \qquad C = 74.$$

Verification of the numerical bounds. We have $3k_1 \in \{i-2, i-1, i\}$ and $3k_2 \in \{2i-2, 2i-1, 2i\}$. The first bound in (12) is equivalent to $3k_1 \leq i$. For the second,

$$5k_2 \geq \frac{5(2i-2)}{3} \geq 3i$$

when $i \geq 10$, which is equivalent to $h_{k_2} \geq 3/2$. For the third,

$$10k_1 \geq \frac{10(i-2)}{3} \geq 3(i-1)$$

when $i \geq 11$, giving the third bound. The fourth follows from $3k_2 \geq 2i - 2$. Finally, $k_2 \leq 2i/3 \leq i - 2$ when $i \geq 6$. The constants in the error bounds are $12 = 3 \cdot 4$ in (11), $(\frac{10}{3} + \frac{3}{2}) \cdot 12 = 58$ in the window sums, and $C = 58 + 16 = 74$ in the final step. $\square$

Proof of Corollary 11. Set $\varepsilon = n^{-p}$, $N = n-1$, $I = \{12, \dots, n-1\}$, and $m = |I| = N - 11$.

For the lower bound, fix any ε-accurate head with $d_h \le r$. For all sufficiently large n, every $i \in I$ has $i\varepsilon \le n^{1-p} \le 10^{-3}$, so Theorem 10 gives $\max_j |A(0)_{ij} - T(i,j)| \le 74\varepsilon$; since $74\, i\varepsilon \le 74\, n^{1-p} < 1/2$ for large n, Lemma 2 applies to row i of $A(0)$ at entrywise error 74ε and gives on I the uniform margins

$$\delta_i \le \log \frac{1 + 74\, i\varepsilon}{1 - 74\, i\varepsilon} \le 222\, n^{1-p} =: \delta, \qquad \gamma_i \ge \log \frac{1 - 74\, i\varepsilon}{74\, i\varepsilon} \ge (p-1)\log n - \log 222 =: \gamma,$$

using $\log \frac{1+x}{1-x} \le 3x$ and $1 - x \ge \frac{1}{3}$ at $x = 74\, i\varepsilon \le 0.074$. The all-zero score matrix $S(0)$ satisfies $\rho(S(0)) \le r$ and $B(S(0)) \le B_\infty$. The lower-bound calculation in the proof of Theorem 5, with N replaced by m and B by B_∞, therefore gives

$$\log B_\infty \ge \frac{m}{r}\big((p-1)\log n + \log\log n + O_{p,r}(1)\big) + O_p(\log n).$$

Replacing $m = N - 11$ by N changes only the remainder, and taking the infimum over feasible heads proves the lower bound.

For the upper bound, take $S_\star$ from Theorem 4; its statement gives $\operatorname{rank}(S_\star) = \rho(S_\star) = r$. Choose a rank factorization $S_\star = UV^\top$, let $a_j, b_j \in \mathbb{R}^r$ be the j-th rows of U, V, and set

$$d_h = r, \quad d = 2r, \quad e_j = (a_j, b_j), \quad u_j = 0, \quad Q = [I_r \;\; 0], \quad K = [0 \;\; I_r].$$

Every payload realizes $S(v)_{ij} = a_i^\top b_j = S_\star(i,j)$: the head is content-independent and ε-accurate, with $B_\infty = B(S_\star)$. Theorem 4, evaluated at $\varepsilon = n^{-p}$ as in the proof of Theorem 5, gives the matching upper estimate. $\square$

The polynomial radius budget of Corollary 7 transfers to the adaptive model.

Corollary 14 (Adaptive radius budget)**.** *Fix $p > 3/2$ and $c > 0$. An ε-accurate head with $\varepsilon = n^{-p}$ and $B_\infty \le n^c$ has*

$$d_h \ge \left(\frac{p - 3/2}{c + p - 1} - o(1)\right) n.$$

Proof. Set $\varepsilon = n^{-p}$, $N = n-1$, $I = \{12, \dots, n-1\}$, and $m = |I| = N - 11$. As in the proof of Corollary 11, for all sufficiently large n, Theorem 10 and Lemma 2 give on I the uniform margins

$$\delta = 222\, n^{1-p}, \qquad \gamma = (p-1)\log n - \log 222,$$

and the all-zero score matrix satisfies $\rho(S(0)) \le d_h$ and $B(S(0)) \le B_\infty \le n^c$.

If $d_h \ge m$, the conclusion is immediate because $(p - 3/2)/(c + p - 1) < 1$ and $m/n \to 1$. Suppose instead that $d_h < m$. Eventually $\gamma \ge \delta$, so Proposition 13 gives $2B_\infty \ge \gamma \ge \delta$ and

$$\gamma^m \le 2^{m-1}(\delta\sqrt{m})^{m-d_h}(2B_\infty\sqrt{m})^{d_h},$$

and therefore

$$d_h \log \frac{2B_\infty}{\delta} \ge m \log \frac{\gamma}{\delta\sqrt{m}} - (m-1)\log 2.$$

Uniformly in d_h,

$$\log \frac{\gamma}{\delta\sqrt{m}} = (p - 3/2)\log n + \log\log n + O_p(1), \qquad \log \frac{2B_\infty}{\delta} \le (c + p - 1)\log n + O_{p,c}(1).$$

The quantity $m\log(\gamma/(\delta\sqrt{m})) - (m-1)\log 2$ is positive for large n. Replacing the multiplier of d_h by its upper bound, dividing by $m\log n$, and using $m/n \to 1$ proves the claim. The remainders depend only on p, c, so the resulting $o(1)$ is uniform in d_h. $\square$

F Experimental details

The offline executable audit is `experiments/run_one_head_experiments.py`; it writes `experiments/output/one_head_bracket.csv`, `experiments/output/one_head_anchor.csv`, and `experiments/output/one_head_thresholds.csv`, together with `experiments/output/environment.txt`.

The manuscript source, the three audit programs, their committed outputs, and `C17/OneHead.lean` are available at `https://anonymous.4open.science/r/rank-logit-law`.

The bracket grid uses $n \in \{12, 16, 20, 24, 32, 40, 48, 56, 64\}$ and $r \in \{1, 2, 4, 8\}$. The program constructs rank-one block factors from binary64-generated budgets. High-precision decimal arithmetic recomputes each worst-row error and centered radius. The program also solves (5) and evaluates (7). Decimal precision follows the proved budget with 80 guard digits.

The threshold audit generates the representability table and the head-dimension crossings at $n = 8192$. Its length searches record the last admitted value followed by the first excluded value. Dimension searches record the last excluded value followed by the first admitted value. The environment file identifies the interpreter, platform, Decimal exponent limit, and truncation cutoff used to produce the artifacts. Each search records its two-sided decision margin; the smallest reported is 2.3×10^{-4} nats. A second program, `experiments/certify_one_head_thresholds.py` recomputes every reported crossing and the $n = 8192$ anchor bound with integer Stirling numbers and outward-rounded interval arithmetic, certifying each decision in `experiments/output/one_head_interval_certificates.csv`. The program `experiments/verify_finite_inequalities.py` checks the permutation count of Lemma 12 exhaustively through $m = 8$ and inequalities (5), (6), and the Hadamard bound directly on every bracket instance.

A Lean 4 development accompanies the project. For the statements of this paper it machine-checks the centered-rank-one, constant-profile specialization of Theorem 3: with the causal-difference profile taken as a hypothesis, $\gamma^N \leq 2B\,\delta^{N-1}$ for every positive boundary count N (`C17/OneHead.lean`, building against Mathlib alone). The development builds with no `sorry` and no project axiom, and its theorems depend only on `propext`, `Classical.choice`, and `Quot.sound`. The general row-subset inequality, the permutation count of Lemma 12, the Hadamard bound, and the construction of Theorem 4 are not formalized; their finite-instance verification is the executable audit above.